INCH-POUND

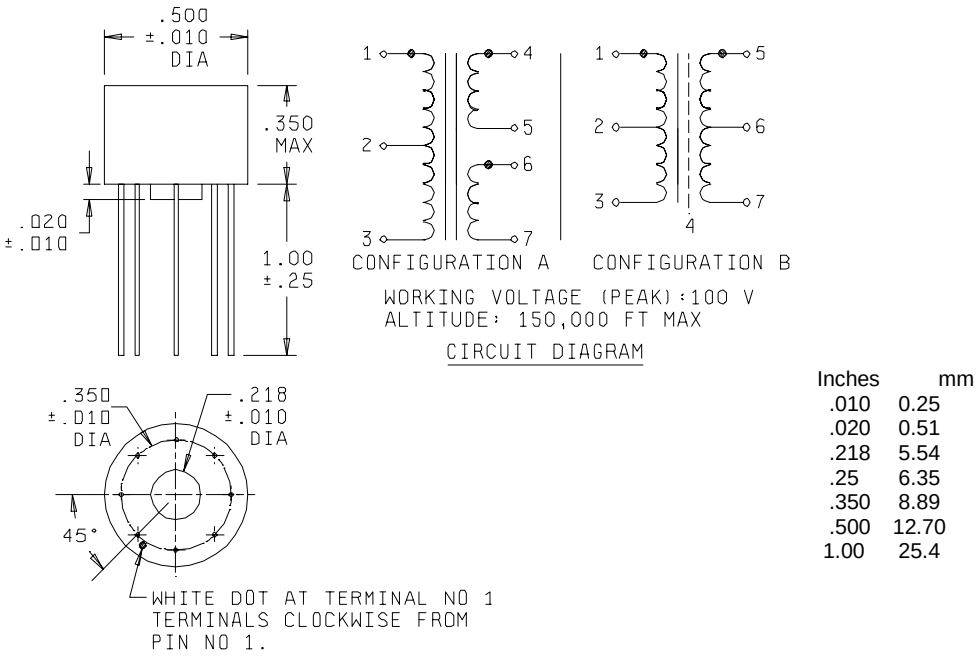
MIL-PRF-27/357C
5 September 2002
SUPERSEDING
MIL-PRF-27/357B
14 June 1988

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, AUDIO FREQUENCY, ULTRA-MINIATURE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-27.




| Inches | mm |
|---|---|
| .010 | 0.25 |
| .020 | 0.51 |
| .218 | 5.54 |
| .25 | 6.35 |
| .350 | 8.89 |
| .500 | 12.70 |
| 1.00 | 25.4 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on the side of the case.
4. Primary and secondary electrical values, specified in table I, shall be marked as applicable.
5. Dots on circuit diagram indicate polarity.

FIGURE 1. Dimensions and configuration.


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Electrical ratings: See table I.

Working voltage (peak): 100 volts.

Frequency range: 300 Hz to 75 kHz.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Terminals: Solid wire.

Composition: Copper clad steel.

Diameter: 0.016 inch ±0.001 (0.41 ±0.03 mm).

Length: 1.00 inch ±0.25 (25.4 ±6.35 mm).

Final finish: Tinned, 100 percent, electroplated in accordance with ASTM B 545.

Weight: 5 grams max.

Altitude: 150,000 feet, maximum.

Operating temperature range: -55°C to +130°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 2.0 pounds.

Thermal shock: 25 cycles, method 107, MIL-STD-202, test condition A-1.

Vibration (high frequency): MIL-STD-202, method 204, test condition B.

Dielectric withstanding voltage:

At sea level: 200 V rms.

At barometric pressure: 200 V rms.

Electrical characteristics:

No load (center tap unbalanced): ±1 percent at 1 volt, 5 kHz across the primary.

Harmonic distortion: Total harmonic content of output shall be a maximum of 5 percent at the specified power level (see table I).

Insertion loss: At the specified power level (see table I), the insertion loss shall be a maximum of 3.0 dB at 1 kHz.

Frequency response: ±3 dB at the rated source and load impedances (see table 1) with 1.0 mW level output and a reference frequency of 1 kHz over a frequency range of 300 Hz to 75 kHz.

Polarity: Additive when appropriate terminals are connected. For configuration A, connect terminals 3 and 4, 5 and 6. For configuration B, connect terminals 3 and 5 (see figure 1, configurations A and B).

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TABLE I. Electrical ratings. 1/

| Dash no. | Primary impedance (ohms) | Secondary Impedance (ohms) | | Primary unbalanced dc current (mA) | Primary dc resistance (ohms) | Secondary dc resistance (ohms) | | Turns ratio Primary:secondary | | Circuit diagram configura-tion |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Series connec-tion | Parallel connec-tion | | | Series connec-tion | Parallel connec-tion | Secondary series connection | Secondary parallel connection | |
| 001 | 20 ct | 8 ct | 2 | 30.0 | 2 | 1.2 | .3 | 1.58:1 | 3.16:1 | A |
| 002 | 20 ct | 20 ct | 5 | 30.0 | 2 | 2.8 | .7 | 1:1 | 2:1 | " |
| 003 | 32 ct | 32 ct | 8 | 23.0 | 3.3 | 4.5 | 1.13 | 1:1 | 2:1 | " |
| 004 | 40 ct | 40 ct | 10 | 21.0 | 3.7 | 5 | 1.25 | 1:1 | 2:1 | " |
| 005 | 48 ct | 16 ct | 4 | 19.0 | 3.75 | 3 | .75 | 1.73:1 | 3.46:1 | " |
| 006 | 50 ct | 50 ct | 12.5 | 19.0 | 5 | 7 | 1.75 | 1:1 | 2:1 | " |
| 007 | 80 ct | 32 ct | 8 | 15.0 | 7.6 | 4.3 | 1.13 | 1.58:1 | 3.16:1 | " |
| 008 | 100 ct | 40 ct | 10 | 13.0 | 10.8 | 5 | 1.25 | 1.58:1 | 3.16:1 | " |
| 009 | 100 ct | 100 ct | 25 | 13.0 | 10.8 | 12 | 3 | 1:1 | 2:1 | " |
| 010 | 120 ct | 12.8 ct | 3.2 | 12.0 | 9.3 | 2.2 | .55 | 3.06:1 | 6.12:1 | " |
| 011 | 120 ct | 60 ct | 15 | 12.0 | 9 | 7.2 | 1.8 | 1.41:1 | 2.83:1 | " |
| 012 | 120 ct | 120 ct | 30 | 12.0 | 11.6 | 13 | 3.3 | 1:1 | 2:1 | " |
| 013 | 150 ct | 12 ct | 3 | 11.0 | 11.2 | 2.2 | .55 | 3.54:1 | 7.08:1 | " |
| 014 | 150 ct | 50 ct | 12.5 | 11.0 | 13 | 6.8 | 1.7 | 1.73:1 | 3.46:1 | " |
| 015 | 150 ct | 100 ct | 25 | 11.0 | 13 | 11.8 | 3 | 1.22:1 | 2.45:1 | " |
| 016 | 150 ct | 150 ct | 37.5 | 11.0 | 13 | 18.3 | 4.6 | 1:1 | 2:1 | " |
| 017 | 150 ct | 600 ct | 150 | 11.0 | 13 | 75 | 19 | 1:2 | 1:1 | " |
| 018 | 200 ct | 200 ct | 50 | 10.0 | 19 | 21.3 | 5.4 | 1:1 | 2:1 | " |
| 019 | 300 ct | 12 ct | 3 | 8.0 | 24 | 1.7 | .43 | 5:1 | 10:1 | " |
| 020 | 300 ct | 50 ct | 12.5 | 8.0 | 30 | 6.8 | 1.7 | 2.45:1 | 4.9:1 | " |
| 021 | 300 ct | 150 ct | 37.5 | 8.0 | 30 | 18 | 4.6 | 1.41:1 | 2.83:1 | " |
| 022 | 300 ct | 300 ct | 75 | 8.0 | 30 | 34 | 8.5 | 1:1 | 2:1 | " |
| 023 | 300 ct | 600 ct | 150 | 8.0 | 30 | 60 | 15 | 1.1:41 | 1.41:1 | " |
| 024 | 320 ct | 12.8 ct | 3.2 | 7.5 | 31 | 1.7 | .43 | 5:1 | 10:1 | " |
| 025 | 400 ct | 40 ct | 10 | 7.0 | 41 | 4.5 | 1.13 | 3.16:1 | 6.23:1 | " |
| 026 | 400 ct | 120 ct | 30 | 7.0 | 41 | 12 | 3 | 1.83:1 | 3.65:1 | " |
| 027 | 400 ct | 200 ct | 50 | 7.0 | 41 | 21.3 | 5.4 | 1.41:1 | 2.83:1 | " |
| 028 | 400 ct | 400 ct | 100 | 7.0 | 41 | 37.5 | 9.4 | 1:1 | 2:1 | " |
| 029 | 400 ct | 1000 ct | 250 | 7.0 | 41 | 96 | 24 | 1:1.58 | 1.26:1 | " |
| 030 | 400 ct | 2000 ct | 500 | 7.0 | 41 | 212 | 53 | 1:2.24 | 1:1.12 | " |
| 031 | 400 ct | 4000 ct | 1000 | 7.0 | 41 | 375 | 94 | 1:3.16 | 1:1.58 | " |
| 032 | 500 ct | 12.8 ct | 3.2 | 6.0 | 48 | 1.7 | .43 | 6.25:1 | 12.5:1 | " |
| 033 | 500 ct | 16 ct | 4 | 6.0 | 48 | 1.9 | .5 | 5.6:1 | 11.2:1 | " |
| 034 | 500 ct | 50 ct | 12.5 | 6.0 | 39 | 6.8 | 1.7 | 3.16:1 | 6.32:1 | " |
| 035 | 500 ct | 150 ct | 37.5 | 6.0 | 48 | 18 | 4.6 | 1.83:1 | 3.65:1 | " |

See footnotes at end of table.



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TABLE I. Electrical ratings – Continued. 1/

| Dash no. | Primary impedance (ohms) | Secondary impedance (ohms) | | Primary unbalanced dc current (mA) | Primary dc resistance (ohms) | Secondary dc resistance (ohms) | | Turns ratio Primary:secondary | | Circuit diagram configura-tion |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Series connec-tion | Parallel connec-tion | | | Series connec-tion | Parallel connec-tion | Secondary series connection | Secondary parallel connection | |
| 036 | 500 ct | 500 ct | 125 | 6.0 | 48 | 54 | 13.5 | 1:1 | 2:1 | A |
| 037 | 500 ct | 500 ct | --- | 6.0 | 48 | 68 | --- | 1:1 | --- | B 2/ |
| 038 | 500 ct | 600 ct | 150 | 6.0 | 48 | 59 | 15 | 1:1.1 | 1.83:1 | A |
| 039 | 600 ct | 12.8 ct | 3.2 | 5.5 | 53 | 1.7 | .43 | 6.85:1 | 13.7:1 | " |
| 040 | 600 ct | 30 ct | 7.5 | 5.5 | 53 | 4.2 | 1.1 | 4.47:1 | 8.94:1 | " |
| 041 | 600 ct | 48 ct | 12 | 5.5 | 53 | 6.8 | 1.7 | 3.54:1 | 7.07:1 | " |
| 042 | 600 ct | 200 ct | 50 | 5.5 | 53 | 21.3 | 5.3 | 1.73:1 | 3.46:1 | " |
| 043 | 600 ct | 600 ct | 150 | 5.5 | 57 | 79 | 20 | 1:1 | 2:1 | " |
| 044 | 600 ct | 600 ct | --- | 5.5 | 53 | 75 | --- | 1:1 | --- | B 2/ |
| 045 | 600 ct | 1200 ct | 300 | 5.5 | 53 | 105 | 26 | 1.1:41 | 1.41:1 | A |
| 046 | 640 ct | 12.8 ct | 3.2 | 5.5 | 66 | 1.7 | .43 | 7.07:1 | 14.14:1 | " |
| 047 | 700 ct | 100 ct | 25 | 5.0 | 69 | 12 | 3 | 2.65:1 | 5.29:1 | " |
| 048 | 800 ct | 12.8 ct | 3.2 | 5.0 | 75 | 1.7 | .43 | 7.9:1 | 15.8:1 | " |
| 049 | 800 ct | 48 ct | 12 | 5.0 | 75 | 6.8 | 1.7 | 4.08:1 | 8.16:1 | " |
| 050 | 800 ct | 800 ct | 200 | 5.0 | 75 | 105 | 26 | 1:1 | 2:1 | " |
| 051 | 900 ct | 600 ct | 150 | 4.5 | 79 | 75 | 19 | 1.22:1 | 2.45:1 | " |
| 052 | 1000 ct | 12 ct | 3 | 4.5 | 103 | 1.6 | .4 | 9.13:1 | 18.3:1 | " |
| 053 | 1000 ct | 50 ct | 12.5 | 4.5 | 103 | 6.4 | 1.5 | 4.47:1 | 8.94:1 | " |
| 054 | 1000 ct | 1000 ct | 250 | 4.5 | 103 | 115 | 29 | 1:1 | 2:1 | " |
| 055 | 1060 ct | 12.8 ct | 3.2 | 4.0 | 108 | 1.7 | .43 | 9.1:1 | 18.2:1 | " |
| 056 | 1200 ct | 12.8 ct | 3.2 | 4.0 | 113 | 1.7 | .43 | 9.68:1 | 19.4:1 | " |
| 057 | 1200 ct | 500 ct | 125 | 4.0 | 113 | 54 | 13.5 | 1.55:1 | 3.1:1 | " |
| 058 | 1200 ct | 1200 ct | 300 | 4.0 | 113 | 126 | 31.6 | 1:1 | 2:1 | " |
| 059 | 1200 ct | 1200 ct | --- | 4.0 | 111 | 125 | --- | 1:1 | --- | B 2/ |
| 060 | 1500 ct | 12 ct | 3 | 3.5 | 126 | 1.6 | .4 | 11.2:1 | 22.4:1 | A |
| 061 | 1500 ct | 500 ct | 126 | 3.5 | 126 | 68 | 17 | 1.73:1 | 3.46:1 | " |
| 062 | 1500 ct | 600 ct | --- | 3.5 | 126 | 94 | --- | 1.58:1 | --- | B 2/ |
| 063 | 1500 ct | 1500 ct | --- | 3.5 | 126 | 176 | --- | 1:1 | --- | B 2/ |
| 064 | 1600 ct | 12.8 ct | 3.2 | 3.5 | 130 | 1.7 | .43 | 11.2:1 | 22.4:1 | A |
| 065 | 2000 ct | 8 ct | 2 | 3.0 | 210 | 1.2 | .3 | 15.8:1 | 31.6:1 | " |

See footnotes at end of table



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TABLE I. Electrical ratings – Continued. 1/

| Dash no. | Primary impedance (ohms) | Secondary impedance (ohms) | | Primary unbalanced dc current (mA) | Primary dc resistance (ohms) | Secondary dc resistance (ohms) | | Turns ratio Primary:secondary | | Circuit diagram configur-ation |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Series connec-tion | Parallel connec-tion | | | Series connec-tion | Parallel connec-tion | Secondary series connection | Secondary parallel connection | |
| 066 | 2000 ct | 2000 ct | 500 | 3.0 | 198 | 218 | 55 | 1:1 | 2:1 | A |
| 067 | 2000 ct | 2000 ct | --- | 3.0 | 198 | 218 | --- | 1:1 | --- | B 2/ |
| 068 | 2000 ct | 8000 ct | 2000 | 3.0 | 198 | 850 | 213 | 1:2 | 1:1 | A |
| 069 | 2500 ct | 2500 ct | 625 | 2.5 | 220 | 305 | 76 | 1:1 | 2:1 | " |
| 070 | 3000 ct | 3000 ct | 750 | 2.5 | 243 | 335 | 84 | 1:1 | 2:1 | " |
| 071 | 3000 ct | 3000 ct | --- | 2.5 | 304 | 335 | --- | 1:1 | --- | B 2/ |
| 072 | 3600 ct | 3600 ct | 900 | 2.5 | 330 | 366 | 92 | 1:1 | 2:1 | A |
| 073 | 4000 ct | 16 ct | 4 | 2.0 | 350 | 3 | .75 | 15.8:1 | 31.6:1 | " |
| 074 | 4000 ct | 600 ct | 150 | 2.0 | 350 | 76 | 19 | 2.58:1 | 5.16:1 | " |
| 075 | 4000 ct | 1000 ct | 250 | 2.0 | 350 | 115 | 29 | 2:1 | 4:1 | " |
| 076 | 4000 ct | 8000 ct | --- | 2.0 | 350 | 1062 | --- | 1:1.41 | --- | B 2/ |
| 077 | 5000 ct | 500 ct | 125 | 2.0 | 350 | 67 | 17 | 3.16:1 | 6.23:1 | A |
| 078 | 5000 ct | 5000 ct | 1250 | 2.0 | 390 | 538 | 135 | 1:1 | 2:1 | " |
| 079 | 7500 ct | 50 ct | 12.5 | 1.5 | 600 | 7 | 1.75 | 12.2:1 | 24.5:1 | " |
| 080 | 7500 ct | 100 ct | 25 | 1.5 | 600 | 12 | 3 | 8.66:1 | 17.3:1 | " |
| 081 | 7500 ct | 600 ct | 150 | 1.5 | 600 | 78 | 19.5 | 3.54:1 | 7.07:1 | " |
| 082 | 7500 ct | 2500 ct | 625 | 1.5 | 600 | 305 | 77 | 1.73:1 | 3.46:1 | " |
| 083 | 7500 ct | 7500 ct | --- | 1.5 | 750 | 828 | --- | 1:1 | --- | B 2/ |
| 084 | 8000 ct | 12.8 ct | 3.2 | 1.5 | 750 | 1.7 | .43 | 25:1 | 50:1 | A |
| 085 | 8000 ct | 1000 ct | 250 | 1.5 | 750 | 115 | 29 | 2.83:1 | 5.66:1 | " |
| 086 | 8000 ct | 1200 ct | --- | 1.5 | 750 | 158 | --- | 2.58:1 | --- | B 2/ |
| 087 | 9000 ct | 9000 ct | --- | 1.4 | 796 | 1100 | --- | 1:1 | --- | B 2/ |
| 088 | 10,000 ct | 12.8 ct | 3.2 | 1.4 | 1048 | 1.7 | .43 | 28:1 | 56:1 | A |
| 089 | 10,000 ct | 16 ct | 4 | 1.4 | 1048 | 3 | .75 | 25:1 | 50:1 | " |
| 090 | 10,000 ct | 100 ct | 25 | 1.4 | 839 | 12 | 3 | 10:1 | 20:1 | " |
| 091 | 10,000 ct | 200 ct | 50 | 1.4 | 839 | 27 | 6.7 | 7.07:1 | 14.14:1 | " |
| 092 | 10,000 ct | 500 ct | 125 | 1.4 | 850 | 67 | 17 | 4.47:1 | 8.94:1 | " |
| 093 | 10,000 ct | 600 ct | 150 | 1.4 | 850 | 74 | 18.5 | 4.08:1 | 8.16:1 | " |
| 094 | 10,000 ct | 1000 ct | 250 | 1.4 | 850 | 115 | 29 | 3.16:1 | 6.32:1 | " |
| 095 | 10,000 ct | 1200 ct | 300 | 1.4 | 850 | 158 | 39.5 | 2.89:1 | 5.77:1 | " |

See footnotes at end of table.





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TABLE I. Electrical ratings - Continued. 1/

| Dash no. | Primary impedance (ohms) | Secondary impedance (ohms) | | Primary unbalanced dc current (mA) | Primary dc resistance (ohms) | Secondary dc resistance (ohms) | | Turns ratio Primary:secondary | | Circuit diagram configur-ation |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Series connec-tion | Parallel connec-tion | | | Series connec-tion | Parallel connec-tion | Secondary series connection | Secondary parallel connection | |
| 096 | 10,000 ct | 1500 ct | 375 | 1.4 | 850 | 177 | 44 | 2.58:1 | 5.16:1 | A |
| 097 | 10,000 ct | 2000 ct | 500 | 1.4 | 850 | 255 | 64 | 2.24:1 | 4.47:1 | " |
| 098 | 10,000 ct | 5000 ct | 1250 | 1.4 | 850 | 630 | 158 | 1.41:1 | 2.83:1 | " |
| 099 | 10,000 ct | 7500 ct | 1875 | 1.4 | 850 | 772 | 193 | 1.15:1 | 2.3:1 | " |
| 100 | 10,000 ct | 10,000 ct | 2500 | 1.4 | 866 | 1215 | 304 | 1:1 | 2:1 | " |
| 101 | 10,000 ct | 10,000 ct | --- | 1.4 | 1060 | 1215 | --- | 1:1 | --- | B 2/ |
| 102 | 15,000 ct | 400 ct | 100 | 1.0 | 1305 | 47 | 12 | 6.12:1 | 12.24:1 | A |
| 103 | 15,000 ct | 500 ct | 125 | 1.0 | 1305 | 53 | 13 | 5.48:1 | 10.96:1 | " |
| 104 | 15,000 ct | 600 ct | 150 | 1.0 | 1305 | 72.5 | 18 | 5:1 | 10:1 | " |
| 105 | 15,000 ct | 10,000 ct | 2500 | 1.0 | 1305 | 1215 | 304 | 1.22:1 | 2.45:1 | " |
| 106 | 15,000 ct | 15,000 ct | --- | 1.0 | 1305 | 1800 | --- | 1:1 | --- | B 2/ |
| 107 | 15,000 ct | 15,000 ct | 3750 | 1.0 | 1305 | 1800 | 450 | 1:1 | 2:1 | A |
| 108 | 16,000 ct | 1000 ct | 250 | 1.0 | 1343 | 115 | 29 | 4:1 | 8:1 | " |
| 109 | 20,000 ct | 500 ct | 125 | 1.0 | 1560 | 68 | 17 | 6.33:1 | 12.65:1 | " |
| 110 | 20,000 ct | 600 ct | 150 | 1.0 | 1560 | 75 | 19 | 5.77:1 | 11.55:1 | " |
| 111 | 20,000 ct | 800 ct | 200 | 1.0 | 1560 | 108 | 27 | 5:1 | 10:1 | " |
| 112 | 20,000 ct | 1000 ct | 250 | 1.0 | 1560 | 121 | 30.5 | 4.47:1 | 8.94:1 | " |
| 113 | 20,000 ct | 5000 ct | 1250 | 1.0 | 1560 | 672 | 168 | 2:1 | 4:1 | " |
| 114 | 20,000 ct | 10,000 ct | 2500 | 1.0 | 1560 | 1520 | 380 | 1.41:1 | 2.83:1 | " |

1/ Qualification testing and approval to M27/357-043 and M27/357-114 shall be sufficient to grant approval to M27/357-001 through M27/357-114.

2/ Includes electrostatic shield. Shield terminal shall be 4. Voltage ratio shall be 2 to 1 at 20 kHz.





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Self-resonance: The self-resonant frequency shall be greater than 150 kHz when measured with the secondaries carrying the specified load and the secondary voltage observed.

Power output in watts (max): .3 at 300 Hz and higher, .4 at 400 and higher, .6 at 1 kHz and higher.

Marking location: See figure 1.

Part or Identifying Number (PIN): M27/357- (dash number from table I)

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5950-1093)

Review activities:
Army - AR, CR4
Navy - AS, MC, OS
Air Force - 99



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